INSTITUTE OF MATHEMATICAL SCIENCES, FACULTY OF SCIENCE, UNIVERSITI MALAYA
SEMESTER 2, SESSION 2025/2026
SIM1006 - TUTORIAL 2

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1. Determine whether the given equation is separable, linear, neither, or both.

   (a) $x^2\frac{dy}{dx} + \sin x - y = 0$

   (b) $\frac{dx}{dt} + xt = e^x$

   (c) $(t^2+1)\frac{dy}{dt} = yt - y$

   (d) $3t = e^t\frac{dy}{dt} + y\ln t$

   (e) $x\frac{dx}{dt} + t^2x = \sin t$

   (f) $3r = \frac{dr}{d\theta} - \theta^3$

2. Find the general solution of the following differential equation.

   (a) $(x-1)dy + y(y-2)dx = 0.$

   (b) $y' = x^{n-1}(1+y^2)$, n is constant.

3. Solve the following differential equation.

   (a) $2y\,dx + x^2\,dy = -dx, \quad y = \frac{7}{2}$ when $x = \frac{1}{\ln 2}$

   (b) $\sin\theta\,dr = 2r\cos\theta\,d\theta, \quad r\left(\frac{\pi}{4}\right) = -2.$

   (c) $y' = \frac{6y\ln y}{x}, \quad y(1) = e.$

   (d) $y' + 3y(y+1)\sin(2x) = 0, \quad y(0) = 1.$

   (e) $\frac{dy}{dx} = \frac{x^2}{y+x^3y}, \quad y(0) = 2.$

   (f) $y' = y^2 + y - 6, \quad y(5) = 10.$

4. Solve the following differential equations.

   (a) $x' + \frac{3}{y}x = 2y$

   (b) $\frac{dy}{dx} = (e^y - x)^{-1}$

   (c) $x\frac{dy}{dx} + 3(y+x^2) = \frac{\sin x}{x}$

5. Solve the following initial value problem.

   (a) $x^2\frac{dy}{dx} - 2xy = x^4 + 3, \quad y(1) = 1.$

   (b) $\frac{dy}{dx} + 4y - e^{-x} = 0, \quad y(0) = \frac{4}{3}.$

   (c) $\frac{dy}{dx} + \frac{3y}{x} + 2 = 3x, \quad y(1) = 1.$

   (d) $\sin x\frac{dy}{dx} + y\cos x = x\sin x, \quad y\left(\frac{\pi}{2}\right) = 2.$